

Daily Sequences Drill #4 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: *Recurrence Relations · The Characteristic Equation · Closed-Form Solutions.*

Section A Rapid Recognition

1. A recurrence has $a_1 = 2$ and $a_{n+1} = 3a_n - 1$. Find a_2 and a_3 .

Answer: $a_2 = 5$, $a_3 = 14$

Method: $a_2 = 3(2) - 1 = 5$. $a_3 = 3(5) - 1 = 14$.

Investigate further: This is a first-order recurrence with a constant additive term — exactly the shape C1/C2 solve properly in closed form later in this sheet.

2. A recurrence has $a_1 = 1$ and $a_{n+1} = 2a_n + 1$. Find a_4 .

Answer: 15

Method: $a_2 = 2(1) + 1 = 3$. $a_3 = 2(3) + 1 = 7$. $a_4 = 2(7) + 1 = 15$.

Investigate further: Notice the pattern $1, 3, 7, 15, \dots$: each is one less than a power of 2 ($2, 4, 8, 16$) — worth guessing the closed form $a_n = 2^n - 1$ and checking it.

3. Write down the characteristic equation of $a_n = 5a_{n-1} - 6a_{n-2}$.

Answer: $x^2 = 5x - 6$ (i.e. $x^2 - 5x + 6 = 0$)

Method: Replace a_n, a_{n-1}, a_{n-2} with x^2, x^1, x^0 respectively in $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This substitution is purely formal — it works because the recurrence is linear with constant coefficients, so a solution of the form $a_n = x^n$ must satisfy this polynomial exactly.

4. Solve $x^2 - 5x + 6 = 0$ for its roots.

Answer: $x = 2, 3$

Method: $x^2 - 5x + 6 = (x - 2)(x - 3) = 0$.

Investigate further: These are the same two numbers as the recurrence's coefficients 5 (their sum) and 6 (their product) — always true by Vieta's formulas on the characteristic equation.

5. A recurrence has characteristic roots 2 and 3. Write down the general form of its closed-form solution.

Answer: $a_n = A \cdot 2^n + B \cdot 3^n$

Method: Two distinct real roots r_1, r_2 give a general solution $a_n = Ar_1^n + Br_2^n$, with A, B determined

by initial conditions.

Investigate further: This is the recurrence-relation analogue of a second-order linear ODE's general solution — two independent “basis” sequences, combined linearly.

6. The recurrence $a_{n+1} = \frac{1}{2}a_n$, $a_1 = 8$ is secretly which type of sequence from an earlier day? Write its closed form.

Answer: A GP; $a_n = 8\left(\frac{1}{2}\right)^{n-1}$

Method: $a_{n+1} = \frac{1}{2}a_n$ is a constant-ratio recurrence, i.e. exactly a geometric sequence with $a = 8$, $r = \frac{1}{2}$ (Day 3's toolkit).

Investigate further: Recognising this instantly, without reaching for a characteristic equation, is the point of B9/C6 later in this sheet.

7. Find the fixed point L of the recurrence $a_{n+1} = 3a_n + 4$ (i.e. solve $L = 3L + 4$).

Answer: $L = -2$

Method: $L = 3L + 4 \Rightarrow -2L = 4 \Rightarrow L = -2$.

Investigate further: L is the value at which the recurrence would stay put forever if it ever landed exactly on it — the fixed point is the anchor the whole shift-to-GP trick (B2, B3, C1, C2) is built around.

8. True or false: every recurrence $a_n = pa_{n-1} + qa_{n-2}$ with real, distinct characteristic roots has a closed form that's a combination of two exponential terms.

Answer: True

Method: This is exactly the general solution form derived in A5: two distinct real roots give a linear combination of two exponential sequences.

Investigate further: This stops being true the moment the roots repeat or go complex — A10 and C4 cover those two other cases.

9. The sequence $a_n = 2^n + 3^n$ satisfies a recurrence $a_n = pa_{n-1} + qa_{n-2}$. Find p and q .

Answer: $p = 5, q = -6$

Method: Since $a_n = 2^n + 3^n$ is built from roots 2 and 3, its characteristic equation is $(x-2)(x-3) = x^2 - 5x + 6 = 0$, i.e. $x^2 = 5x - 6$, giving $a_n = 5a_{n-1} - 6a_{n-2}$.

Investigate further: This is A3/A4 run in reverse — given the roots (or a sequence built from them), reconstruct the recurrence rather than solve one.

10. A recurrence has a repeated characteristic root $r = 2$. Write down the general form of its closed-form solution.

Answer: $a_n = (A + Bn)2^n$

Method: A repeated root r needs a second, genuinely independent solution alongside r^n — multiplying by n (giving nr^n) supplies it, so the general solution is $a_n = (A + Bn)r^n$.

Investigate further: This mirrors what happens with a repeated root of a polynomial more generally — an extra factor of n (or x) is needed to keep the two basis solutions linearly independent.

Section B Manipulation Drills

1. Solve the recurrence $a_n = 5a_{n-1} - 6a_{n-2}$ with $a_1 = 1, a_2 = 4$, for the constants A, B in $a_n = A \cdot 2^n + B \cdot 3^n$.

Answer: $A = -\frac{1}{2}, B = \frac{2}{3}$

Method: $n = 1$: $2A + 3B = 1$. $n = 2$: $4A + 9B = 4$. Doubling the first: $4A + 6B = 2$; subtracting from the second: $3B = 2 \Rightarrow B = \frac{2}{3}$. Then $2A + 2 = 1 \Rightarrow A = -\frac{1}{2}$.

Investigate further: Always check both original conditions after solving: $a_1 = -\frac{1}{2}(2) + \frac{2}{3}(3) = -1 + 2 = 1 \checkmark$, $a_2 = -\frac{1}{2}(4) + \frac{2}{3}(9) = -2 + 6 = 4 \checkmark$.

2. For $a_{n+1} = 3a_n + 4$, the fixed point is $L = -2$. Letting $b_n = a_n - L = a_n + 2$, what type of sequence is (b_n) ?

- A) A GP with ratio 3.
 B) An AP with common difference 3.
 C) A GP with ratio -2 .
 D) Neither.

Answer: A) A GP with ratio 3.

Method: $b_{n+1} = a_{n+1} + 2 = (3a_n + 4) + 2 = 3a_n + 6 = 3(a_n + 2) = 3b_n$ — so (b_n) satisfies a pure geometric recurrence.

Investigate further: This is exactly why the shift $b_n = a_n - L$ works in general: subtracting the fixed point cancels the additive constant, leaving pure multiplication behind.

3. Continuing from B2: given $a_1 = 1$ (so $b_1 = 3$), find a_5 .

- A) 241
 B) 243
 C) 239
 D) 245

Answer: A) 241

Method: $b_1 = a_1 + 2 = 3$, and (b_n) is a GP with ratio 3: $b_n = 3 \cdot 3^{n-1} = 3^n$. So $b_5 = 3^5 = 243$, and $a_5 = b_5 - 2 = 241$.

Investigate further: Always convert back from b_n to a_n at the very end — forgetting the -2 shift-back is the most common slip in this technique.

4. Find the characteristic equation and its roots for $a_n = a_{n-1} + 2a_{n-2}$.

Answer: $x^2 - x - 2 = 0$; roots $x = 2, -1$

Method: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 = (x - 2)(x + 1) \Rightarrow x = 2, -1$.

Investigate further: A negative characteristic root is completely normal — it produces a sign-alternating term $B(-1)^n$ in the closed form, as B5 uses directly.

5. A recurrence has characteristic roots 2 and -1 , with $a_1 = 1, a_2 = 5$. Which closed form fits?

- A) $a_n = 2^n + (-1)^n$
 B) $a_n = 2^n - (-1)^n$
 C) $a_n = 2 \cdot 2^n - (-1)^n$
 D) $a_n = 2^{n-1} + (-1)^{n-1}$

Answer: A) $a_n = 2^n + (-1)^n$

Method: Check $n = 1$: $2 + (-1) = 1 \checkmark$. Check $n = 2$: $4 + 1 = 5 \checkmark$. The other options fail at $n = 1$: B) gives $2 - (-1) = 3$, C) gives $4 + 1 = 5$, D) gives $1 + 1 = 2$ — none match $a_1 = 1$.

Investigate further: With two initial conditions and two unknowns (A, B in $A \cdot 2^n + B \cdot (-1)^n$), the closed form is uniquely determined — always verify a proposed closed form against *both* given terms, not just one.

6. Which of the following recurrences does NOT have a characteristic equation with two distinct real roots?

- A) $a_n = 5a_{n-1} - 6a_{n-2}$
 B) $a_n = 4a_{n-1} - 4a_{n-2}$
 C) $a_n = a_{n-1} + 2a_{n-2}$
 D) $a_n = 3a_{n-1} - 2a_{n-2}$

Answer: B) $a_n = 4a_{n-1} - 4a_{n-2}$

Method: Its characteristic equation is $x^2 - 4x + 4 = (x - 2)^2 = 0$ — a repeated root, discriminant $16 - 16 = 0$. The other three all have positive discriminant (A: $25 - 24 = 1$; C: $1 + 8 = 9$; D: $9 - 8 = 1$), giving distinct real roots.

Investigate further: Computing the discriminant of the characteristic equation directly ($p^2 + 4q$ for $a_n = pa_{n-1} + qa_{n-2}$) is faster than factorising every option by hand.

7. A recurrence has repeated characteristic root $r = 3$, with $a_1 = 6, a_2 = 27$. Using $a_n = (A + Bn)3^n$, find A and B .

- A) $A = 1, B = 1$
 B) $A = 2, B = 1$
 C) $A = 1, B = 2$
 D) $A = 0, B = 2$

Answer: A) $A = 1, B = 1$

Method: $(A + B \cdot 1) \cdot 3^1 = 6 \Rightarrow A + B = 2$. $(A + B \cdot 2) \cdot 3^2 = 27 \Rightarrow A + 2B = 3$. Subtracting: $B = 1$, then $A = 1$.

Investigate further: Double-check: $a_1 = (1 + 1) \cdot 3 = 6 \checkmark$, $a_2 = (1 + 2) \cdot 9 = 27 \checkmark$.

8. Using the closed form from B7, find a_4 .

Answer: 405

Method: $a_n = (1+n)3^n$. $a_4 = (1+4) \cdot 3^4 = 5 \times 81 = 405$.

Investigate further: The repeated-root closed form grows faster than a plain exponential r^n because of the extra factor of n — compare $a_4 = 405$ here to what a plain $c \cdot 3^4 = 81c$ would need ($c = 5$) to match, which is a coincidence of this specific example, not a general shortcut.

9. A recurrence is defined by $a_{n+1} = a_n + 5$, $a_1 = 2$. Is this recurrence's closed form better found using the AP toolkit (Day 1) or the characteristic-equation toolkit (Day 4)?

- A) The AP toolkit — it's simply $a_n = 2 + 5(n-1)$.
- B) The characteristic-equation toolkit — solve $x = x + 5$.
- C) Both give the same closed form, but the characteristic equation is faster.
- D) Neither toolkit applies.

Answer: A) The AP toolkit — it's simply $a_n = 2 + 5(n-1)$.

Method: A constant additive step (+5 every time, no multiplication) is precisely Day 1's arithmetic sequence definition — the characteristic-equation machinery is unnecessary (and its "fixed point" equation $x = x + 5$ has no solution, signalling this recurrence sits outside that method's normal use case).

Investigate further: Recognising when a lighter toolkit already solves the problem is as important as knowing the heavier one — see Top 5 Pattern 4.

10. A recurrence has characteristic equation $x^2 - 6x + 9 = 0$. What is special about its roots?

- A) Two distinct real roots.
- B) A repeated real root.
- C) Two complex roots.
- D) No real roots.

Answer: B) A repeated real root.

Method: $x^2 - 6x + 9 = (x-3)^2 = 0$ — discriminant $36 - 36 = 0$.

Investigate further: This is the same equation structure as B7's recurrence, $a_n = 6a_{n-1} - 9a_{n-2}$ — worth noticing the connection.

Section C Substitution & Structure

1. Which substitution transforms $a_{n+1} = 4a_n - 3$ into a pure geometric recurrence $b_{n+1} = 4b_n$?

- A) $b_n = a_n - 1$
- B) $b_n = a_n + 1$
- C) $b_n = a_n - 3$
- D) $b_n = a_n/4$

Answer: A) $b_n = a_n - 1$

Method: Fixed point: $L = 4L - 3 \Rightarrow -3L = -3 \Rightarrow L = 1$. So $b_n = a_n - 1$, giving $b_{n+1} = a_{n+1} - 1 = (4a_n - 3) - 1 = 4a_n - 4 = 4(a_n - 1) = 4b_n$.

Investigate further: The fixed point of $a_{n+1} = pa_n + c$ is always $L = \frac{c}{1-p}$ (for $p \neq 1$) — worth memorising as a one-line formula rather than resolving $L = pL + c$ from scratch every time.

2. The recurrence $a_{n+1} = 4a_n - 3$ with $a_1 = 2$ has closed form $a_n = 1 + c \cdot 4^{n-1}$. Find c .

- A) 1
B) 2
C) 3
D) 4

Answer: A) 1

Method: At $n = 1$: $a_1 = 1 + c \cdot 4^0 = 1 + c = 2 \Rightarrow c = 1$.

Investigate further: c here is exactly b_1 from the shift-to-GP method (C1) — the two approaches (“closed form with unknown constant” vs. “shift to a GP”) are the same technique in different notation.

3. A recurrence has characteristic roots $r_1 = 5$ and $r_2 = -2$. Using $p = r_1 + r_2$ and $q = -r_1r_2$, write down the recurrence $a_n = pa_{n-1} + qa_{n-2}$.

- A) $a_n = 3a_{n-1} + 10a_{n-2}$
B) $a_n = 3a_{n-1} - 10a_{n-2}$
C) $a_n = -3a_{n-1} + 10a_{n-2}$
D) $a_n = 7a_{n-1} - 10a_{n-2}$

Answer: A) $a_n = 3a_{n-1} + 10a_{n-2}$

Method: $p = 5 + (-2) = 3$. $q = -(5)(-2) = 10$. Check via the characteristic equation directly: $(x - 5)(x + 2) = x^2 - 3x - 10 = 0 \Rightarrow x^2 = 3x + 10 \Rightarrow a_n = 3a_{n-1} + 10a_{n-2}$.

Investigate further: This is A9’s technique in general form — always rebuild the characteristic equation from the roots directly as a check, rather than trusting the $p = r_1 + r_2$, $q = -r_1r_2$ formulas from memory alone.

4. A recurrence has characteristic equation $x^2 - 2x + 5 = 0$. What can be said about its roots?

- A) Two distinct real roots.
B) A repeated real root.
C) Two complex conjugate roots, $1 \pm 2i$.
D) No roots exist.

Answer: C) Two complex conjugate roots, $1 \pm 2i$.

Method: Discriminant $= (-2)^2 - 4(1)(5) = 4 - 20 = -16 < 0$, so the roots are complex: $x = \frac{2 \pm \sqrt{-16}}{2} = \frac{2 \pm 4i}{2} = 1 \pm 2i$.

Investigate further: A negative discriminant doesn't mean "no solution" for a recurrence the way it might for a real-world root-finding problem — the recurrence still has a well-defined (complex-valued, or real-oscillating) closed form, explored further in D3.

5. Which technique is most appropriate for solving $a_{n+1} = 2a_n + 3^n$ (note the non-constant additive term)?

- A) The constant-shift trick ($b_n = a_n - L$ for a fixed L) — works directly.
- B) The homogeneous characteristic-equation method, plus a particular solution of the form $C \cdot 3^n$ added to the general solution.
- C) Treating it as a plain AP.
- D) It cannot be solved by any method covered so far.

Answer: B) The homogeneous method plus a particular solution $C \cdot 3^n$.

Method: The constant-shift trick only cancels a *constant* forcing term — here the forcing term 3^n grows with n , so no fixed constant L can absorb it. The standard fix is to add a particular solution of the same form as the forcing term ($C \cdot 3^n$) to the homogeneous general solution.

Investigate further: This is the recurrence-relation analogue of "guess a particular solution matching the forcing function's form" from differential equations — the same idea, one level more discrete.

6. Which of the following recurrences is secretly equivalent to an arithmetic sequence — Day 1's toolkit is sufficient, no characteristic equation needed?

- A) $a_{n+1} = a_n + 7$
- B) $a_{n+1} = 2a_n$
- C) $a_{n+1} = 2a_n + 7$
- D) $a_n = 5a_{n-1} - 6a_{n-2}$

Answer: A) $a_{n+1} = a_n + 7$

Method: A) is a constant additive step — an AP, Day 1's toolkit exactly. B) is a constant-ratio step — a GP (Day 3). C) needs the shift-to-GP trick (still not a plain AP). D) genuinely needs the full characteristic equation.

Investigate further: This question is the whole sheet's "triage" skill in miniature: always ask whether Day 1 or Day 3's lighter toolkit already applies before reaching for Day 4's heavier machinery.

7. A recurrence $a_n = pa_{n-1} + qa_{n-2}$ has BOTH characteristic roots equal to 1. What does this imply about a_n as $n \rightarrow \infty$ (assuming (a_n) is not identically zero)?

- A) Exponential growth.
- B) Linear growth in n (or constant) — effectively an AP.
- C) The sequence must be periodic.
- D) The sequence must converge to a finite limit.

Answer: B) Linear growth in n (or constant) — effectively an AP.

Method: A repeated root $r = 1$ gives closed form $a_n = (A + Bn)(1)^n = A + Bn$ — exactly linear in n (an AP) if $B \neq 0$, or constant if $B = 0$.

Investigate further: This is a genuinely elegant full-circle result: the “two independent exponential terms” structure of a general second-order recurrence collapses back down to Day 1’s arithmetic sequence exactly when both roots equal 1 — see D4 for the reverse direction (which (p, q) forces this).

8. A recurrence has characteristic roots r and $\frac{1}{r}$ (reciprocal roots, $r \neq \pm 1$). What can be said about p and q in $a_n = pa_{n-1} + qa_{n-2}$?

- A) $q = -1$ always, while $p = r + \frac{1}{r}$ can be any value with $|p| \geq 2$.
 B) $p = -1$ always.
 C) p and q must both equal 1.
 D) No general statement can be made.

Answer: A) $q = -1$ always, $p = r + \frac{1}{r}$ can be any value with $|p| \geq 2$.

Method: $q = -r_1r_2 = -r \cdot \frac{1}{r} = -1$, independent of r . And $p = r + \frac{1}{r}$, which by AM–GM-style reasoning satisfies $|r + \frac{1}{r}| \geq 2$ for any nonzero real r (with equality only at $r = \pm 1$).

Investigate further: The bound $|r + \frac{1}{r}| \geq 2$ is the same inequality that appears throughout the Algebra pillar’s toolkit — a nice example of one pillar’s tool resurfacing to describe a genuinely different structure (reciprocal characteristic roots) here.

Section D Challenge

1. A sequence is defined by $a_1 = 1, a_2 = 2$, and for $n \geq 3$: $a_n = \frac{a_{n-1} + a_{n-2}}{2}$. Find $\lim_{n \rightarrow \infty} a_n$.

Answer: $\frac{5}{3}$

Method: Characteristic equation: $2x^2 = x + 1 \Rightarrow 2x^2 - x - 1 = 0 = (2x + 1)(x - 1) \Rightarrow x = 1, -\frac{1}{2}$. General solution: $a_n = A + B(-\frac{1}{2})^n$. Using $a_1 = 1$: $A - \frac{B}{2} = 1$. Using $a_2 = 2$: $A + \frac{B}{4} = 2$. Subtracting: $\frac{3B}{4} = 1 \Rightarrow B = \frac{4}{3}$, then $A = 1 + \frac{2}{3} = \frac{5}{3}$. As $n \rightarrow \infty$, $(-\frac{1}{2})^n \rightarrow 0$, so $a_n \rightarrow A = \frac{5}{3}$.

Investigate further: Numerically: 1, 2, 1.5, 1.75, 1.625, 1.6875, ... — oscillating and visibly closing in on $1.6 = \frac{5}{3}$. A root of exactly 1 in an otherwise-decaying recurrence always signals convergence to a finite, computable limit, never to 0.

2. A recurrence satisfies $a_{n+1} = 6a_n - 9a_{n-1}$ (for $n \geq 2$), with $a_1 = 1, a_2 = 6$. Using the repeated-root closed form $a_n = (A + Bn)3^n$, find a_5 .

- A) 405
 B) 243
 C) 324
 D) 486

Answer: A) 405

Method: Characteristic equation $x^2 - 6x + 9 = (x - 3)^2 = 0$ — repeated root 3. Solving $a_n = (A + Bn)3^n$:

$n = 1$: $(A + B)(3) = 1 \Rightarrow A + B = \frac{1}{3}$. $n = 2$: $(A + 2B)(9) = 6 \Rightarrow A + 2B = \frac{2}{3}$. Subtracting: $B = \frac{1}{3}$, so $A = 0$. Thus $a_n = \frac{n}{3} \cdot 3^n = n \cdot 3^{n-1}$. $a_5 = 5 \times 3^4 = 5 \times 81 = 405$.

Investigate further: Check the pattern directly: $a_n = n \cdot 3^{n-1}$ gives 1, 6, 27, 108, 405, ... — verify $a_3 = 6a_2 - 9a_1 = 6(6) - 9(1) = 27$ ✓ matches the recurrence.

3. A recurrence has characteristic equation $x^2 + 4 = 0$ (i.e. $a_n = -4a_{n-2}$). What is true about its behaviour?

- A) The sequence oscillates in sign while its magnitude grows like 2^n — consistent with complex roots of magnitude 2.
- B) The sequence converges to 0.
- C) The sequence is eventually constant.
- D) The sequence is purely real-valued growth with no oscillation.

Answer: A) The sequence oscillates in sign while its magnitude grows like 2^n .

Method: $x^2 = -4 \Rightarrow x = \pm 2i$, purely imaginary roots of magnitude 2. Testing directly with $a_1 = 1, a_2 = 0$: $a_3 = -4a_1 = -4$, $a_4 = -4a_2 = 0$, $a_5 = -4a_3 = 16$, $a_6 = -4a_4 = 0$, $a_7 = -4a_5 = -64$ — the nonzero terms alternate sign and grow by a factor of 4 every two steps (i.e. magnitude 2 per step), exactly matching complex roots of modulus 2.

Investigate further: This is the general pattern for complex characteristic roots $re^{i\theta}$: the real sequence's magnitude grows like r^n , while it oscillates with a “period” related to θ — purely imaginary roots ($\theta = 90$) give the fastest possible sign-oscillation, a full cycle every 4 steps.

4. For which of the following (p, q) pairs does $a_n = pa_{n-1} + qa_{n-2}$ produce a closed form that reduces exactly to Day 1's arithmetic-sequence form (i.e. a_n linear in n), for suitable initial conditions?

- A) $(p, q) = (2, -1)$
- B) $(p, q) = (1, 1)$
- C) $(p, q) = (-2, 1)$
- D) $(p, q) = (3, -2)$

Answer: A) $(p, q) = (2, -1)$

Method: Linear-in- n behaviour needs a repeated root of exactly 1: $(x - 1)^2 = x^2 - 2x + 1 = 0$, giving $p = 2, q = -1$ (matching $x^2 - px - q = 0$). Checking D: $(p, q) = (3, -2)$ gives $x^2 - 3x + 2 = (x - 1)(x - 2) = 0$ — roots 1, 2, distinct, so the general solution $A + B \cdot 2^n$ is linear in n only in the degenerate special case $B = 0$, not for suitable general initial conditions.

Investigate further: This is C7's result read backwards — given the desired long-term behaviour (linear growth), work out which characteristic equation, and hence which (p, q) , produces it.

5. A recurrence has characteristic roots r_1, r_2 with $|r_1| < 1$ and $|r_2| < 1$. What happens to a_n as $n \rightarrow \infty$, regardless of initial conditions?

- A) $a_n \rightarrow 0$ always.
- B) a_n diverges to infinity.

- C) a_n oscillates forever without settling.
 D) It depends on the initial conditions.

Answer: A) $a_n \rightarrow 0$ always.

Method: $a_n = Ar_1^n + Br_2^n$ (or the repeated-root analogue). Since $|r_1| < 1$ and $|r_2| < 1$, both $r_1^n \rightarrow 0$ and $r_2^n \rightarrow 0$ as $n \rightarrow \infty$, regardless of the constants A, B (which are fixed by the initial conditions but don't affect whether the limit is 0).

Investigate further: This is the recurrence-relation version of Day 3's convergence condition $|r| < 1$ for a GP — both roots need to be inside the unit circle in magnitude, not just one. Tomorrow's sheet studies exactly this kind of boundedness and convergence behaviour in general.

Top 5 Patterns — Worksheet #4

- Characteristic equation: turn $a_n = pa_{n-1} + qa_{n-2}$ into $x^2 - px - q = 0$.** The entire toolkit starts here. (see A3, A4, A9, B4, C3)
- Distinct real roots $\rightarrow a_n = Ar_1^n + Br_2^n$; repeated root $\rightarrow a_n = (A + Bn)r^n$.** Always check the discriminant first to know which shape applies. (see A5, A10, B5, B7, C7)
- First-order recurrences $a_{n+1} = pa_n + c$ shift to a pure GP via the fixed point $L = \frac{c}{1-p}$.** Let $b_n = a_n - L$, solve the GP, shift back. (see A7, B2, B3, C1, C2)
- Recognise the degenerate cases first.** Constant-difference recurrences are Day 1's AP; constant-ratio recurrences are Day 3's GP — don't reach for the characteristic equation when a lighter toolkit already solves it. (see A6, B9, C6)
- Complex characteristic roots signal oscillation with exponentially changing magnitude,** not simple convergence or divergence. (see C4, D3)

Common Traps — Worksheet #4

- Forgetting to check for a repeated root before using the two-distinct-roots closed form.** A zero discriminant needs $(A + Bn)r^n$ instead. (see B6, B10)
- Mixing up the sign convention when reading p, q off from given roots.** Remember $x^2 - px - q = 0$, not $x^2 + px + q = 0$. (see C3, D4)
- Applying the constant-shift trick to a recurrence with a non-constant forcing term.** $b_n = a_n - L$ only cancels a fixed constant — it does nothing against a term like 3^n . (see C5)
- Assuming every recurrence converges to a finite limit.** Convergence needs *every* characteristic root to have magnitude strictly less than 1. (see D5)
- Solving for A, B using only one initial condition.** A second-order recurrence's closed form has two free constants and needs two equations to pin them both down. (see B1, B5, B7)

“A repeated root does not give you $Ar^n + Br^n$. It gives you $(A + Bn)r^n$, and the n is not decoration.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths